Russian Geometry

16, 23 December 2023

Problem 1. Let P and Q be points on the sides AB and BC of $\triangle ABC$ such that $AP = CQ$, and let W be the midpoint of $\widehat{ABC}$. Prove that the circumcircle of $\triangle BPQ$ passes through W.

Problem 2. Let P and Q be points on the sides AB and BC of $\triangle ABC$ such that $AP + CQ = AC$ and let I be the incentre of $\triangle ABC$. Prove that the circumcircle of $\triangle BPQ$ passes through I.

Problem 3. Let L be the midpoint of $\widehat{AC}$, not containing B, of the circumcircle of an acute-angled $\triangle ABC$, and let P be the foot of the perpendicular from B to the tangent to the circumcircle at L . Prove that P, L, and the midpoints of sides AB, BC are concyclic.

Problem 4. The diagonals of a rectangle $ABCD$ meet at point E. A circle centered at E lies inside the rectangle. Let CF, DG, AH be the tangents to this circle from C, D, A, and let CF meet DG at point I, EI meet AD at point J, and AH meet CF at point L. Prove that LJ is perpendicular to AD.

Problem 5. A circle touches the lateral sides of a trapezium $ABCD$ at points B and C, and its centre lies on AD. Prove that the diameter of the circle is less than the medial line of the trapezium.

Problem 6. Let D and E be points on the lateral sides AB and BC, respectively, of an isosceles $\triangle ABC$ such that $\angle BED = 3\angle BDE$, and let D be the reflection of D about AC. Prove that the line DE passes through the incentre of $\triangle ABC$.

Problem 7. Let $ABCD$ be a cyclic quadrilateral, let E and F be points on the sides AD and CD, respectively, such that $AE = BC$ and $AB = CF$, and let M be the midpoint of EF. Prove that $\angle AMC = 90°$.

Problem 8. Let A_1, B_1, C_1 be the feet of the altitudes of an acute-angled $\triangle ABC$. The incircle of $\triangle A_1B_1C_1$ touches A_1B_1, A_1C_1, B_1C_1 at points C_2, B_2, A_2, respectively. Prove that the lines AA_2,BB_2, CC_2 are concurrent at a point lying on the Euler line of $\triangle ABC$.

Problem 9. Let A be a fixed point of a circle ω. Let BC be an arbitrary chord of ω passing through a fixed point P. Prove that the nine-point circles of $\triangle ABC$ touch some fixed circle not depending on BC.

Problem 10. Given $\triangle ABC$ with $BC = a$, $CA = b$, $AB = c$ and $a > b > c$, let its incentre I and the points of contact D and E of the incircle with BC and AC, respectively, be marked. Construct a segment with length $a - c$ using only a ruler and drawing at most three lines.

Problem 11. Let E be the point of contact of the incircle of $\triangle ABC$ with side AC, and let P be a point on side AC. Prove that one of the common internal tangent lines to the incircles of $\triangle ABP$ and $\triangle CBP$ passes through E, and that E, P and the incentres of $\triangle ABP$ and $\triangle CBP$ are concyclic.

Problem 12. Suppose the reflection of the orthocentre of $\triangle ABC$ about its circumcentre lies on BC. Let D be the foot of the altitude from A. Prove that D lies on the circle passing through the midpoints of the altitudes of $\triangle ABC$.

Problem 13. Altitudes BE and CF of an acute-angled $\triangle ABC$ meet at point H. The perpendicular from H to EF meets the line ℓ passing through A and parallel to BC at point P. The bisectors of two angles between ℓ and HP meet BC at points S and T. Prove that the circumcircles of $\triangle ABC$ and $\triangle PST$ are tangent to each other.

Problem 14. Let H be the orthocentre of an acute-angled $\triangle ABC$, let E, F be points on AB, AC respectively, such that $AEHF$ is a parallelogram, let X, Y be the common points of the line EF and the circumcircle ω of $\triangle ABC$, and let Z be the antipode of A in ω. Prove that H is the orthocentre of $\triangle XYZ$.

Problem 15. Given $\triangle ABC$ with obtuse $\angle ABC$, let P and Q be points on AC such that $AP = PB$ and $BQ = QC$. The circumcircle of $\triangle BPQ$ meets the sides AB and BC at points N and M respectively.

(a) Prove that the distances from the common point R of PM and NQ to A and C are equal.

(b) Let BR meet AC at point S. Prove that MN is perpendicular to OS, where O is the circumcentre of $\triangle ABC$.

Problem 16. The base AD of a trapezium $ABCD$ is twice its base BC, and angle C equals one and a half of angle A. The diagonal AC divides angle C into two angles. Which of them is greater?

Problem 17. Let P be a point inside an acute $\angle ABC$, let I_1 and I_2 be the incentres of $\triangle PAB$ and $\triangle PBC$, respectively, and let BL_1 and BL_2 be the angle bisectors of $\angle ABP$ and $\angle CBP$, respectively, where L_1 lies on PA and L_2 lies on PC. Prove that the lines I_1I_2, L_1L_2 and the external bisector of $\angle APC$ are concurrent.

Problem 18. Let $ABCD$ be a convex quadrilateral. Points X and Y lie on the extensions beyond D of the sides CD and AD, respectively, such that $DX = AB$ and $DY = BC$. Similarly, points Z and T lie on the extensions beyond B of the sides CB and AB, respectively, such that $BZ = AD$ and $BT = DC$. Let M_1 be the midpoint of XY, and let M_2 be the midpoint of ZT. Prove that the lines DM_1, BM_2, and AC are concurrent.

Problem 19. Let AH_A and BH_B be the altitudes of $\triangle ABC$. The line H_AH_B meets the circumcircle of $\triangle ABC$ at points P and Q. Let A' be the reflection of A about BC, and B' be the reflection of B about CA. Prove that A', B', P and Q are concyclic.

Problem 20. A common external tangent to circles ω_1 and ω_2 touches ω_1 and ω_2 at points T_1 and T_2, respectively. Let A be an arbitrary point on the extension of T_1T_2 beyond T_1, and let B be a point on the extension of T_1T_2 beyond T_2 such that $AT_1 = BT_2$. The tangents from A to ω_1 and from B to ω_2 distinct from T_1T_2 meet at point C. Prove that all nagelians of $\triangle ABC$ from C have a common point.

Problem 21. Restore a bicentric quadrilateral $ABCD$ if the midpoints of $\overset{\frown}{AB}$, $\overset{\frown}{BC}$, $\overset{\frown}{CD}$ of its circumcircle are given.

Problem 22. Let $ABCD$ be a cyclic quadrilateral. An arbitrary circle passing through C and D meets AC and BC at points X and Y, respectively. Find the locus of common points of circumcircles of $\triangle CAY$ and $\triangle CBX$.

Problem 23. Let D be a point on the median AM of $\triangle ABC$. The tangents to the circumcircle of $\triangle BDC$ at points B and C meet at point K. Prove that DD' is parallel to AK, where D' is the isogonal conjugate of D in $\triangle ABC$.

Problem 24. Let $ABCD$ be a cyclic quadrilateral, let M_{ac} be the midpoint of AC, let H_d and H_b be the orthocentres of $\triangle ABC$ and $\triangle ADC$, respectively, and let P_d and P_b be the projections of H_d and H_b to BM_{ac} and DM_{ac}, respectively. Define P_a, P_c for the diagonal BD in a similar manner. Prove that P_a, P_b, P_c and P_d are concyclic.

Problem 25. Let $\triangle ABC$ be scalene, let M be the midpoint of BC, let P be the point of intersection of AM and the incircle of $\triangle ABC$ closest to A, and let Q be the point of intersection of the ray AM and the excircle farthest from A. The tangent to the incircle at P meets BC at point X, and the tangent to the excircle at Q meets BC at Y. Prove that $MX = MY$.